

Problems Technology Solves: Technical Worksheet

Grade 6 • Worksheet • About 25 minutes

Work with pen and paper. Use color pencils if you like. You do not need a computer or the internet.

Name: _____

Grade/Section: _____

Date: _____

How to use this worksheet

- Do the parts **in order** from top to bottom.
- Each part has a **time** so you know how long to spend.
- Read every instruction before you write. You can do this on your own.

Part 1 – What you will learn (about 3 minutes)

What you will learn: **Technology** is built to fix **problems**—the gap between how things are and how we want them to be. In this worksheet you will name kinds of problems technology tackles, match problems to tools, see how solutions work as **input** → **process** → **output**, and use a simple **solve-it** process while noticing **constraints** (limits like power, cost, or accuracy) and **tradeoffs**.

Do this:

1. For each item below, circle **Problem** or **Solution**.
 - A road is flooded and cars cannot pass: _____
 - A bridge that lets people cross safely: _____
 - Sorting 500 names by hand takes all weekend: _____
2. **Name one problem** in your school or home that a tool (any technology) helps with.

Part 2 – Kinds of problems technology tackles (about 5 minutes)

Technology often helps when work is **too slow or repetitive**, places are **too far or hard to navigate**, there is **too much information to sort**, something is **unsafe or unhealthy**, or people need to **communicate across distance**.

Do this: Match each example to one category from the word bank. Write the category phrase.

Word bank: *Too slow or repetitive · Too far or hard to navigate · Too much information to sort · Unsafe or unhealthy · Hard to communicate across distance*

Example	Your answer (one phrase from the word bank)
A factory robot welds the same joint hundreds of times	
A map app shows the fastest route to a new hospital	
A spreadsheet totals hundreds of sales rows in seconds	
A smoke alarm beeps when it senses smoke	
A video call lets a grandparent see a birthday party far away	

Part 3 – From problem to tool (about 5 minutes)

Good designers start with the **problem**, not the gadget. They ask: *What is wrong? Who needs help? What would “better” look like?*

Real-life scenario: A river town floods often. Engineers place **sensors** in the water and connect them to **alert systems**. When the water rises fast, phones and sirens warn people to move to higher ground. The **problem** is dangerous, sudden flooding. The **technology** helps people get warning in time.

Discuss with a partner or reflect in writing: What problem is being solved? What might people do without this tool?

Do this: For each case, write the **problem** and one **technology** that helps.

Case	Problem (what is wrong?)	Technology that helps
You are lost in a new city and need directions	_____	_____
A club must track who paid dues from 80 members	_____	_____
A person in a wheelchair needs a door to open without turning a knob	_____	_____
Two teammates speak different languages in a chat	_____	_____

Part 4 – How solutions work (about 5 minutes)

Many systems follow **input** → **process** → **output**. **Input** is what goes in (a button press, a sensor reading). **Process** is the rules or steps inside (software, a chip, gears). **Output** is what comes out (a sound, a message, movement). Some systems use **feedback**: the output is checked and the process adjusts (like a thermostat turning heat on or off).

Real-life scenario: A package has a **barcode**. At each stop, a worker **scans** it (input). The shipping computer **updates** the route (process). Your tracking page **shows** “Out for delivery” (output).

Critical thinking: Why might the same tool solve one problem but fail another? Think about accuracy, power, cost, or user error.

Do this: Fill in the table.

Problem	Main part of the system	One way it could fail
Room is too cold at night		
Teacher takes attendance with ID cards		
You need to know if your bus is late		

Part 5 – A simple solve-it process (about 5 minutes)

You can solve problems with a short process:

- 1. **Define the problem** — say what is wrong and for whom.
- 2. **List options** — include tech and non-tech ideas.
- 3. **Pick and test** — try one option; see if it really helps.
- 4. **Improve** — fix what did not work and try again.

Real-life scenario: Items keep getting lost in a classroom: lunch boxes, jackets, and headphones. The class defines the problem: “Things look the same and get mixed up.” They list options: labeled cubbies, a checklist, a simple numbering system, or an app to log items. They test labeled cubbies first and add a sign-out sheet for borrowed headphones.

Discuss with a partner or reflect in writing: When is a low-tech fix better than an app? Give one example.

Do this: Use the four steps for this problem: *Students forget which homework is due each week.*

Step	Your answer
1. Define the problem	_____
2. List two options (tech and/or non-tech)	_____
3. Pick one to test and why	_____
4. One way to improve if it fails	_____

Optional: Sketch a simple diagram of your chosen solution in the box below.

Part 6 – Check what you learned (about 2 minutes)

Checklist. Put a ✓ in the box when it is true for you.

- ☐ I can explain that technology is built to address problems.
- ☐ I can name at least three kinds of problems technology often tackles.
- ☐ I can match a problem to a tool that helps solve it.
- ☐ I understand input → process → output (and feedback) in simple systems.
- ☐ I can use the solve-it steps and name one constraint or tradeoff.

When all boxes are checked, you have reviewed the main ideas from this worksheet.

Worksheet on Problems Technology Solves. Central topic: How technology addresses real-world problems. Main technology area: Problem solving / Systems thinking.